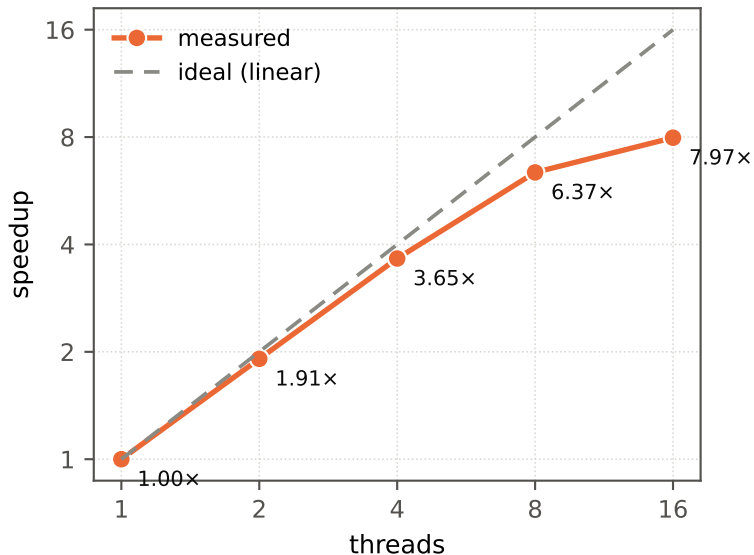
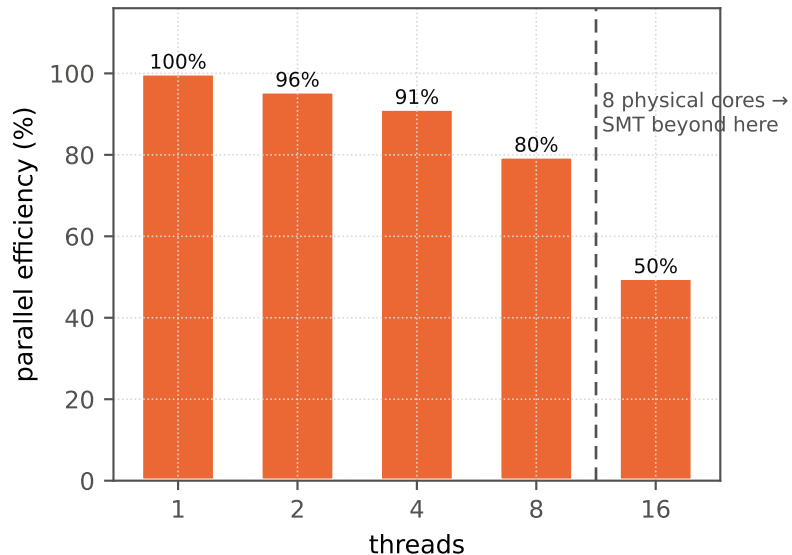


Parallel speedup



Efficiency



Mandelbrot 1920×1080, max_iter=1000, rayon. Efficiency holds above 90% to 4 threads and 80% at 8 — the Ryzen 7 5800H's physical core count — then halves at 16. That is the SMT signature: sibling threads share execution units, so the last doubling buys 25% rather than another 2x.